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Instruments and Experiments
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Understanding Placement-Dependent Frequency Response Variability in a Portable Loudspeaker

Abstract

This experiment investigates the effects of physical location between a sound source and receiver on the frequency response measurement in a reverberant environment. Our research question asks how moving a speaker closer to a microphone and further from a wall would affect the interference patterns caused by reverberation off the walls in a room and, thus, the system's frequency response. To test this, we used a microphone to record an exponential sine sweep played aloud with a portable speaker. We placed the speaker against a wall and moved it away from that wall and towards the microphone in 3-centimeter increments, thus decreasing the distance between the speaker and microphone while increasing the distance from the wall behind. We then processed these recordings by isolating the impulse response using an inverse filter and applying a Fast Fourier Transform (FFT) to generate the magnitude of the frequency response and plot the results, making sure to calibrate the microphone. Our results demonstrate that as the speaker moved away from the wall, the low-frequency bass boost decreased in a physically predictable manner, while mid- and high-frequency responses changed irregularly due to the complex interaction of reflections from all surfaces in the room. These findings confirm that room acoustics introduce position-dependent distortions far exceeding the ±3 dB tolerances commonly reported by speaker manufacturers, emphasizing the importance of measurement conditions when interpreting loudspeaker frequency response specifications.

I. Introduction

1.1 Motivation

Measuring accurate frequency response data is important for designing speakers, rooms with favorable acoustics, and for audio engineering in general. Specifically, the frequency response tells us if a speaker plays music exactly as the artist recorded it or if it distorts the true sound. The frequency response data can be used by engineers to correct these issues and find flaws in speaker hardware. Certain pitches being too loud or too quiet could be telling of a design flaw and call for changes in materials in the speaker cone, a change in the cabinet shape, or some other design adjustment.

In a studio setting, it is important that a producer knows that the audio they are hearing is accurate to avoid making adjustments which worsen the audio all in an effort to maintain the highest quality. In an idealized environment, like an anechoic chamber, you only hear the audio produced by the speaker, making it easy to isolate audio issues as flaws in the speaker design. However, in a non-idealized, real-room environment, sound reverberates off the walls, floor, and ceiling which makes it difficult to determine if the source of undesired noise is the room or the speaker. These reverberations create "comb filtering", as the reflected sound arrives slightly late to the listener and cancels out specific frequencies, leaving a jagged response which misrepresents the speaker's performance. Without the ability to separate the direct and reverberation sounds, flaws may be identified in a speaker that don't actually exist, resulting in unnecessary repairs which waste time and money. Particularly in a home studio environment, people may not have perfectly anechoic rooms, so it is important to understand how to test that a speaker is working properly in such environments and find the favorable position for a speaker and recording device to capture the best audio possible.

These principles are used in Architectural Acoustics fields to design concert halls or offices by measuring how sounds travels and bounces off the walls to determine the best placement for sound absorbent panels to optimize the audio and ensure a speaker or performer's voice from a specific position can be heard from all positions of the audience. Understanding how sound interacts with walls or large equipment can help design solutions for noise mitigation.

1.2 Technical Background

The propagation of sound in fully enclosed spaces is governed by the room acoustics and the signal received by the microphone is the superposition of the direct sound, the early reflections, and the reverberant field. The direct sound takes the shortest path from the source (speaker) to the receiver (microphone) and carries the most accurate representation of the source signal. Nearby surfaces, like the walls, create the early reflections, which arrive shortly after the direct sound followed by the reverberant field, which have a longer delay since they have been diffused by multiple scattering events. The combination of these effects alters the measured signal received by the microphone and distorts the true response frequency.[1]

An important term to describe these delayed signals is comb filtering, which occurs when the direct sound and reflected signals combine with different path lengths. The difference in path lengths introduces a phase shift between the signals, which leads to either constructive or destructive interference of the signals at different frequencies. Comb filtering effects produce peaks and dips in the frequency response, the spacing of which are inversely proportional to the delay between the direct and reflected signals. Thus, reflections from nearby surfaces can significantly distort measurements even with minute changes in geometry.[2]

These effects can be quantified using the direct-to-reverberant ratio (DRR), or the ratio of energy in the direct sound to that in the reverberant field. As the distance between the source and receiver increases, the DRR decreases since the direct sound energy decays with distance while the reverberant energy stays relatively constant throughout the room. Thus, a high DRR is indicative of a measured signal dominated by the direct sound and is therefore a more accurate representation of the source while a low DRR has reduced clarity due to reverberation.[1]

Impulse response measurement techniques are commonly used to analyze such acoustic effects, with one widely used method being the exponential sine sweep. This method employs a sine wave with exponentially increasing frequency that is played through the source which is then convolved with the inverse filter to recover the system's impulse response. This method is advantageous because it produces a cleaner signal with less noise, is less affected by distortion in the system, and can separate distortion effects from the true response.[2, 3, 4]

Once obtained, the impulse response can be transformed to the frequency domain using the Fast Fourier Transform (FFT), which decomposes the signal in the time domain into its constituent frequency components. By representing the impulse response in the frequency domain, features such as resonances, attenuation, and comb filtering effects can be visualized, thus better showing how room acoustics influence the signal.[3, 4]

II. Experimental Methods and Microphone Calibration

To measure the speaker's frequency response, the exponential sine sweep method was used. Since measuring the speaker's output from a recording of music or speech is difficult to analyze clearly, we instead play a specific test signal through the speaker, record the response of the room to that signal, and

mathematically determine the speaker's frequency response from the recording. The specific process is elaborated in this section.

2.1 Generating the Test Signal

The test signal played through the speaker is a sine wave with continuously increasing frequency from 20 to 20,000 Hz over the course of 5 seconds. This frequency range is the entirety of the audible spectrum for humans. The sine sweep is exponential because the rate of increase of the frequency is not constant but exponential; the signal spends equal amounts of time in each octave. This is because human hearing is roughly logarithmic, so spending equal amounts of time in each octave excites all parts of the audible spectrum equally.

Additionally, a companion signal called the inverse filter is generated, which is constructed to mathematically "undo" the exponential sine sweep mentioned previously. It is the sine sweep played in reverse with an additional amplitude envelope compensating for the fact that the lower frequencies in the sweep carry more energy than the higher frequencies. These two signals together can be used to determine the impulse response for any system through which the sweep is played.

Both of these signals were generated using the Generate.m code provided in Appendix A and saved as uncompressed audio files at a sample rate of 48 kHz to ensure the 20 Hz to 20 kHz range is completely captured. That is to say one second of audio is represented by 48,000 individual measurements of the amplitude, providing a high resolution frequency signal to play through the speaker.

2.2 Recording the Room's Response

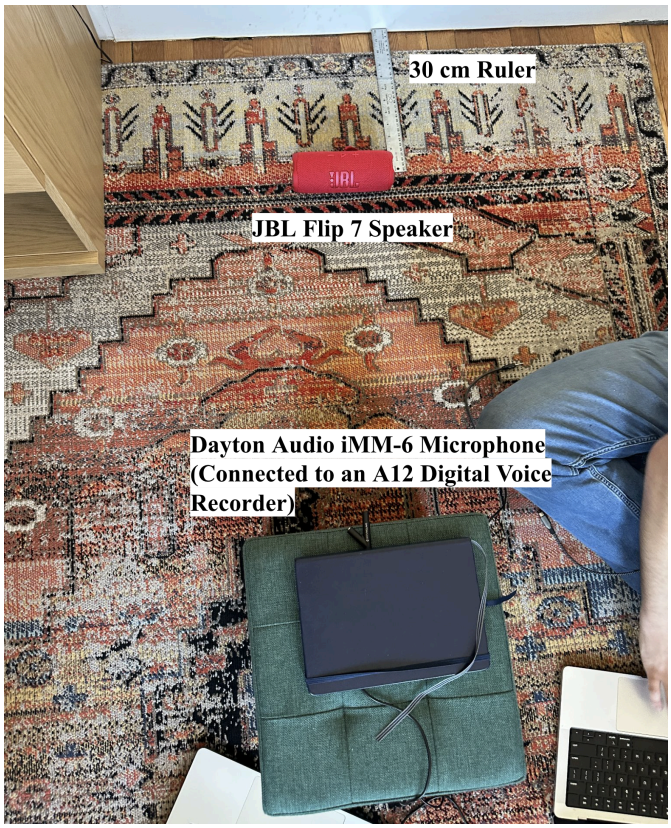


Fig 1. Experimental setup showing the JBL Flip 7 speaker placed incremental distances away from the wall and towards the Dayton Audio iMM-6 microphone connected to an A12 Digital Voice Recorder.

In order to record the response of the room, our experimental setup consisted of a JBL flip speaker to produce the sine sweep, an Dayton Audio iMM-6 microphone (with stereo plug) to record the room's response, an A12 Digital Voice Recorder to cleanly start and stop and house the audio recordings, a 30-centimeter ruler to measure speaker distance from the wall, and various objects to hold the microphone in a fixed position.

For the data collection, the speaker starts directly against the wall and is moved in 3-centimeter increments away from the wall and towards the microphone positioned approximately 1 meter away from the wall. At each distance, the exponential sine sweep is played through the speaker and the response of the room is recorded through the microphone. With this method, the microphone not complying captures the direct sound produced by the speaker, but also the reverberated sound from the speaker reflected off the walls, floor, and ceiling of the room, all of which arrive at the microphone at slightly different times. The combination of the direct and reverberated sounds is the room's response to the sine sweep.

2.3 Extracting the Impulse Response via Deconvolution

Following data collection, the audio recordings were transferred via cable to our computer, loaded into MATLAB, and processed using the code in Appendix B. Here, the signal is first combined with the inverse filter via a convolution operation, which systematically combines every point in the room response signal with every point in the inverse filter signal via the following mathematical formula for continuous signals f(t) and g(t):

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)\,d\tau$$

In other words, the formula calculates the time-domain superposition of the two signals by performing a point-by-point multiplication and integration for every possible relative shift. It effectively combines every point in one signal with every point in the other. Convolving the room's response with the inverse filter results in the room's impulse response, or how the room would respond to an instantaneous and infinitely brief burst of sound. This is because the sweep and inverse filter are designed such that convolving them effectively cancels out the known characteristics of the sweep itself, leaving behind only the contribution of the room.

To account for inconsistencies in the recordings for when the sine sweep begins and making sure all the recordings align temporally, the MATLAB code automatically locates a large spike which occurs at the moment the direct sounds from the speaker arrives at the microphone and discards everything before it. The large spike in the signal is followed by a decaying tail representing the quieter reverberations of the room which arrive at the microphone at different times. The impulse response is then trimmed to be 0.5 seconds to maintain consistency between the signals while capturing the meaningful early reverberations and discarding the later, extremely quiet ones.

2.4 Converting to the Frequency Domain

After the impulse responses are isolated and trimmed for each distance, they are converted from the time-domain into the frequency domain for more meaningful analysis. This is done using the Fast Fourier Transform (FFT), which decomposes a given signal into its constituent sine waves, which reveals how much energy there is at each frequency. The FFT is a high-speed algorithm used to compute the Discrete Fourier Transform (DFT) defined for a sequence of N samples x[n] as:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

The FFT produces a set of complex numbers since the basis functions of the transform, $e^{-j\theta}$, are derived from Euler's Formula, $e^{-j\theta} = \cos(\theta) - j\sin(\theta)$. The magnitude of each complex X[k] is taken to find the intensity of the signal. This results in a raw, uncalibrated frequency response curve expressed in volts for each signal, with each "bin", k, corresponding to a specific frequency.

2.5 Calibrating the Microphone

The raw output of the FFT is in volts, the electrical unit of the microphone's output which reflects both the true room response to the speaker and the microphone's non-uniform sensitivity across frequencies. Calibration is used to correct the latter and ensure the frequency response reflects the true room impulse response in the frequency domain.

This calibration is specific to the microphone itself and was found by searching the manufacturer's website using the microphone's unique serial number.[5] This file contains the frequency-dependent correction values in decibels (dB) and specifies how much the microphone over- or under-responds to a 1 kHz reference level. This calibration data is loaded into MATLAB in the file Calibration.txt.

Since the FFT produces results in evenly spaced frequency "bins" that do not match the frequencies in the calibration file, linear interpolation is used in the code to compute the exact correction ended at each frequency bin of the FFT output. The code then adds these corrections directly to the magnitude values after they are converted to decibels using the following formula:

$$Magnitude(dB) = 20\log_{10}(|X[k]|)$$

After calibration, the frequency response reflects the speaker's true acoustic output in physically meaningful, decibel units.

2.6 Smoothing and Plotting

The resulting calibrated frequency impulse response is a jagged curve at fine frequency resolution as a result of interference dips caused by reverberation in the room and due to random noise in the microphone measurements. To better visualize trends in the response, Gaussian smoothing is applied in the code, which replaces each data point with a weighted average of neighboring data points. So, points closest to the center of the smoothing window, in our case, 800 frequency bins, are weighted more heavily than those at the edges. This smoothing technique maintains the dips that occur due to physical interference from reverberation and only removes background noise that would make the signal curves difficult to compare.

The resulting smoothed curves for the frequency-domain impulse response at every position of the speaker are plotted on the same set of axes. Frequency is plotted from 40 Hz to 4 kHz on a logarithmic scale which, as mentioned previously, is roughly consistent with human perception of pitch.

III. Results and Analysis

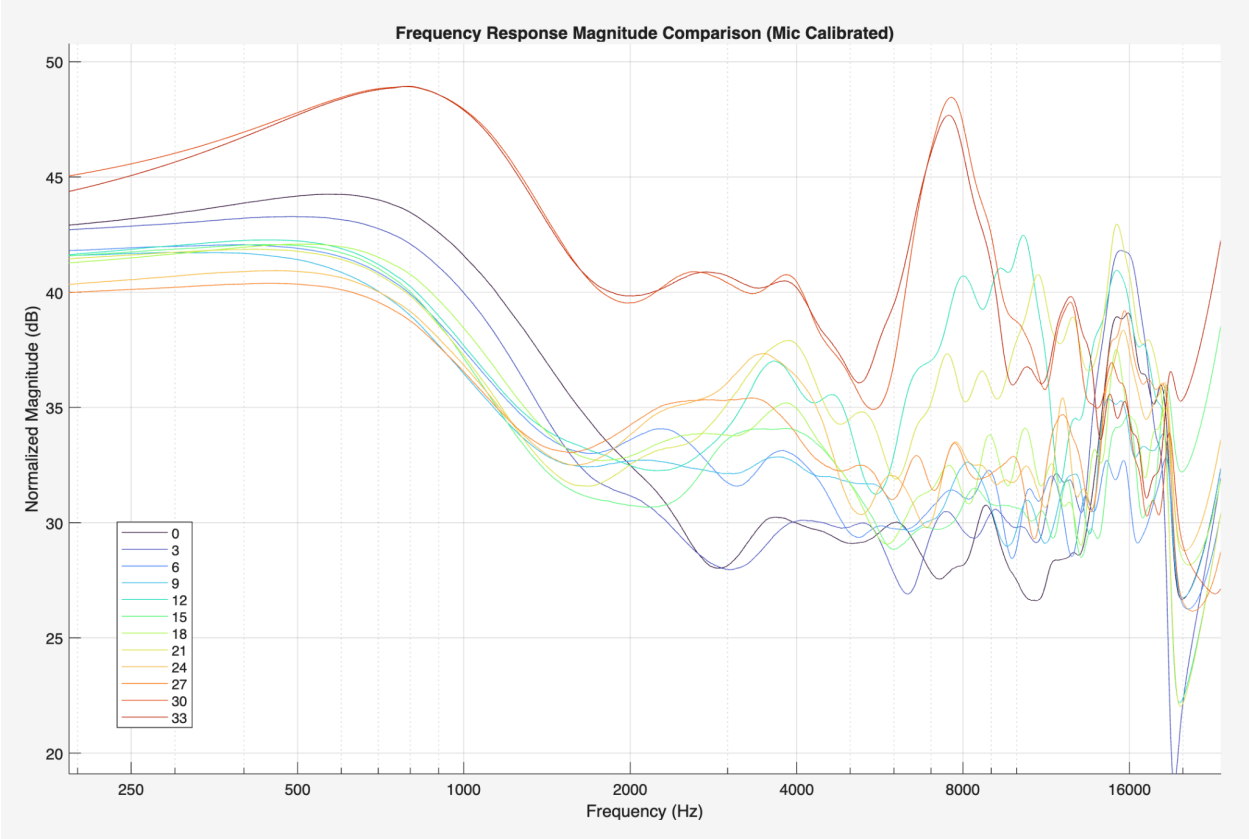


Fig 2. Calibrated, smoothed frequency impulse response magnitude curves for various speaker positions ranging from 0 cm to 33 cm from the wall.

3.1 Analysis of Trends in the Frequency Response

Figure 2 shows the calibrated and smoothed frequency impulse response magnitude curves for all twelve speaker positions, ranging from 0 to 33 cm from the wall in 3 cm increments. Each curve represents the speaker's measured output, in dB, across the audible frequency range, from approximately 200 Hz to 20 kHz.

The most consistent trend feasible in the figure can be seen in the low frequency region (<1 kHz), where the curves separate the most clearly by position. At 0 cm from the wall, the bass response peaks at approximately 50 dB near 700 Hz, while subsequent curves have roughly decreasing peaks with distance from the wall. This is explained by the fact that at closer distances to the wall, the speaker uses the wall as a reflector and reinforces low frequency output by preventing sound from radiating behind the speaker. However, as the speaker moves away from the wall from 0 to 27cm, the bass peak roughly decreases with each step. This trend is physically consistent with the comb filtering framework described in the introduction. When the speaker is directly against the wall, the sound reflected off the walls travels only a short distance before reaching the microphone, arriving almost simultaneously with the direct sound. These two copies of the sound add constructively at low frequencies, amplifying the pass. However, as distance from the wall increases, the delay of the reverberated signal increases, meaning that the two signals are less synchronized and add more destructively.

There is a notable departure from this trend for the 30 and 33 cm distances, as both curves are observed to be significantly above the other distances. This is most likely due to experimental error, such

as uncontrolled or inconsistent changes in the speaker or microphone position between trials. The majority of the results for the 3 cm increments demonstrate that small changes in position are enough to impart significant changes in the frequency response. This is particularly the case at higher frequencies where wavelengths are short. It is also possible that there was a change in volume or orientation of the speaker or in the surrounding environment that affected the acoustic conditions.

Above around 2 kHz, the curves become significantly more jagged and show less systematic trends associated with speaker position. Since wavelengths are shorter at higher frequencies, even variations in the microphone or speaker position at the millimeter-scale are enough to produce changes in the interference pattern between direct and reverberant sounds. Thus, the high frequency response is more sensitive to incidental positing errors than the low frequency response and the spread between curves reflects experimental variability more so than the effects of wall distance. However, the higher-frequency curves do exhibit increase in complexity with more peaks of greater depth at frequencies above 8 kHz. This makes sense due to the comb filtering effect becoming more pronounced at high frequencies, where sound waves are shorter and more sensitive to small changes in the source and receiver positions.

3.2 Comments on Noise Levels

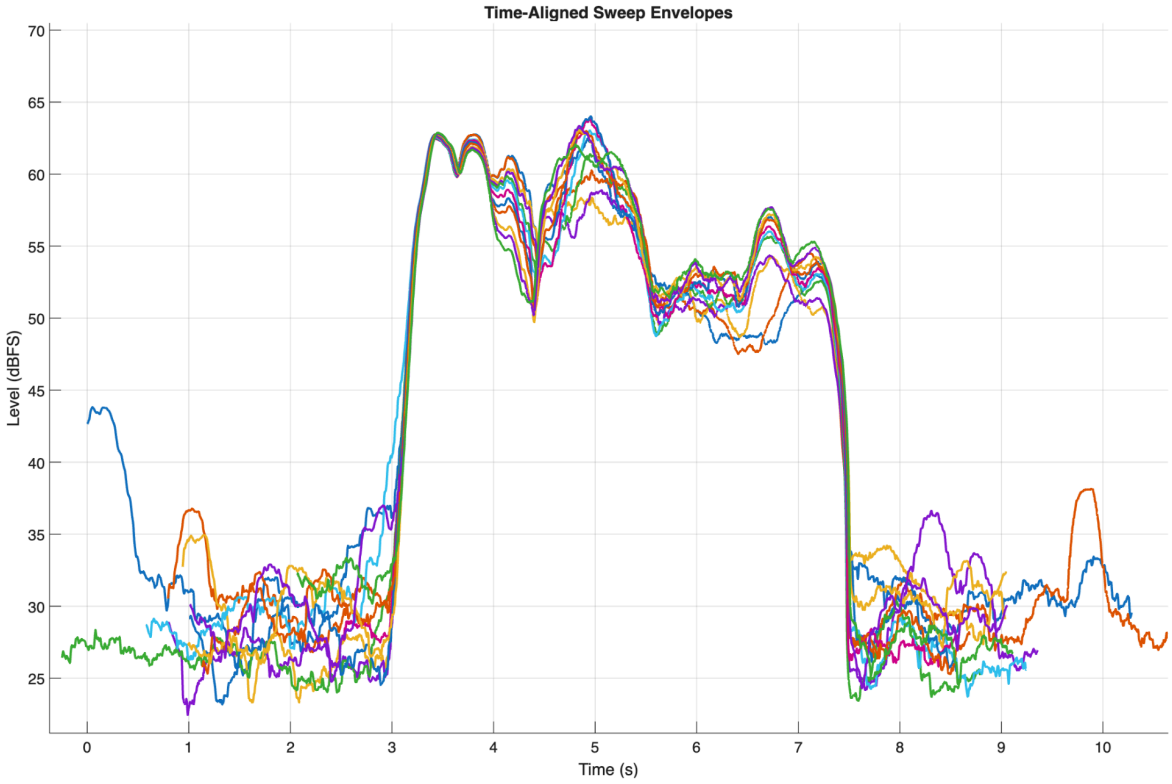


Fig 3. Signal amplitude for each sine sweep recording over time (including before and after) demonstrating the high signal-to noise ratio. (Appendix C)

To verify that background noise did not meaningfully contaminate the measurements, the signal-to-noise ratio (SNR) of the recording was assessed by comparing the measured signal level of approximately 50 dB to the background noise floor of approximately 30 dB, seen in Figure 3, yielding an SNR of 20 dB. To quantify the practical impact of this noise floor on the measurement, the noise

contribution can be subtracted from the signal in the power domain using the expression $10 \cdot \log(10^{50/10} - 10^{20/10})$, which yields approximately 49.996 dB, a difference of less than 0.005 dB from the raw signal level. This is far below the just-noticeable difference (JND) for loudness in human hearing, which is approximately 1 dB under typical listening conditions. As a result, error due to background noise was deemed negligible and no noise correction was applied to the data in post-processing.

IV. Conclusions and Future Work

Overall, the most important observation in our results is the highly nonlinear relationship between the speaker distance from the wall and the resulting frequency response. Unlike a speaker measured in an anechoic chamber where moving the microphone would produce a smooth, predictable change in level, the curves in Figure 2 do not change proportionally as the speaker moves in equal increments. This is because the frequency response is shaped not only by the wall behind the speaker, but by reflections from every surface in the room, each traveling a different path length with a different delay to the microphone. Small changes in speaker position shifts all these path lengths at once, altering the interference of some frequencies but not others and producing irregularly evolving curves.

4.1 Implications for Speaker Frequency Response Specifications

From the perspective of speaker design, our results show how manufacturers often intentionally shape the frequency response of their speakers to emphasize bass frequencies, artificially boosting the low end to make the speaker sound more impressive. This is visualized in Figure 2 by the peaks in the frequency responses for all distances in the 600 - 700 Hz range.

A striking feature of Figure 2 is the spread of the magnitude levels between the different positions across the full frequency range. Across all twelve positions, the measured magnitude at any given frequency varies by as much as ±15 dB. Industry convention is to report whether a speaker's frequency response is within a ±3 dB tolerance, or a 6 dB window across its rated frequency range. This is used as the threshold for a flat, desirable response where all frequencies are equally represented in magnitude by the speaker's output. This specification is typically measured under controlled anechoic conditions with the microphone placed at a single, optimized position directly on-axis with the speaker.

The data in Figure 2 demonstrates that this specification can be very misleading to consumers who are unfamiliar with the subtleties of acoustic measurement. Simply moving the speaker across a 33 cm range within a room without changing any other variables produces variation far exceeding the ±3 dB tolerance advertised by manufacturers. In a reverberant environment, a speaker's frequency response is not a fixed property of the device but rather a property of the device in combination with the room it is placed in and its precise position in that room. A flat frequency response measured in an anechoic chamber would be expected to remain roughly flat regardless of microphone placement since there are no reflections to create the position-dependent interference. This stability does not exist in a reverberant room. Thus, reporting a ±3 dB specification derived from anechoic measurements without disclosing the measurement conditions can significantly misrepresent real-world performance.

4.2 Future Work

There are various areas of this project we would like to explore in more depth. We would like to run the study again with much smaller distance increments as our measurements did not provide enough granularity. We would also like to run the experiment in different sized rooms to see if the variation we observed is specific to the environment we tested in, or indicative of general loudspeaker behavior in

rooms. We are also interested in comparing experimentally collected frequency response data to simulated frequency response data in a virtually designed room using software like COMSOL or Rew.

References

[1] D. Griesinger, "The importance of the direct to reverberant ratio in the perception of distance, localization, clarity, and envelopment, Part two.," The Journal of the Acoustical Society of America 125(4_Supplement), 2483–2483 (2009).

[2] S. Müller, and P. Massarani, "Transfer-Function Measurement with Sweeps," Journal of the Audio Engineering Society 49, (2001).

[3] A. Farina, "Advancements in Impulse Response Measurements by Sine Sweeps," Journal of The Audio Engineering Society, (2007).

[4] A. Farina, "Simultaneous Measurement of Impulse Response and Distortion With a Swept-Sine Technique," (2000).

[5] "Dayton Audio Microphone Calibration Tool"
https://support.daytonaudio.com/MicrophoneCalibrationTool

AI Statement

We used AI assistance Level 2 to write our report. More specifically, we began by having AI write an outline for our paper and then wrote the paper ourselves but prompted AI throughout the writing process to answer questions. In the outline, the AI provided bullet points of salient background information to include and a general outline of what information to include in each section of the paper based on the guidelines. As we were writing, since we were learning a lot of new concepts related to speakers, reverberated signals, and audio signal processing that we weren't familiar with before, Claude specifically was a helpful tool in explaining concepts in simpler terms than the scientific papers we encountered. However, in prompting AI during this process, paraphrasing or using AI written sentences was somewhat unavoidable, making our process somewhere between a level 2 and level 4 overall.

For the code specifically, we wrote basic functional code that worked and used AI, Gemini specifically, to add more features (like grabbing many files from a folder simultaneously and color coding the plots). We also used AI to gain an understanding of how the math used by the code works. Though the AI did not write this code itself, it provided useful background information to enhance our understanding and better explain the code in the report.